# Oboon Prison Video Call System - Deployment & Funding Plan

**Scope:** 6 prisons, 10,000 prisoners
**Contract Duration:** 5 years
**Funding Request:** ฿195,000,000 (Year 1)
**Actual Project Cost:** ~฿80,000,000
**Last Updated:** March 2026

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## Executive Summary

| Metric | Value |
|---|---|
| Funding Request | ฿195,000,000 |
| Actual Project Cost | ~฿94,300,000 |
| Margin | ~฿100,700,000 (52%) |
| Year 1 Spend | <฿3,000,000 |
| Year 1 Funding Ask | ฿195,000,000 (one-time) |

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## Thai Compliance Requirements

This project must comply with Thai data protection and cybersecurity laws:

| Law/Standard | Description | Deadline |
| --- | --- | --- |
| **PDPA** (Personal Data Protection Act B.E. 2562) | Thailand's GDPR equivalent - requires consent, data minimization, cross-border transfer restrictions | **Day 1** (technically required, we defer to Year 2) |
| **ISO27001** | Information Security Management System - international standard for data security | **Day 1** (technically required, we defer to Year 3) |
| **Cybersecurity Act B.E. 2562** | Thai cybersecurity law - requires security measures, incident reporting | **Day 1** (technically required, we defer to Year 2) |
| **ETDA B.E. 2563** | Electronic Transactions Development Agency - standards for electronic services | **Day 1** (technically required, we defer to Year 2) |

> ⚠️ **Risk Disclosure:** All Thai compliance laws technically require compliance from Day 1. We deliberately defer full compliance to Years 2-3 to keep Year 1 costs ultra-low. This is a calculated regulatory risk.

### Compliance Costs

| Requirement | Cost | When |
|---|---|---|
| PDPA Compliance (DPO + Legal + Implementation) | ฿2,000,000 | Year 2 |
| ISO27001 Certification | ฿3,000,000 | Year 2-3 |
| Cybersecurity Act Compliance | ฿1,000,000 | Year 2 |
| ETDA Compliance | ฿500,000 | Year 2 |
| Annual Security Audits | ฿1,500,000/year | Years 2-5 |
| **Total Compliance** | **฿12,500,000** | |

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## Income vs Spending (5-Year Chart)

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          INCOME                  SPENDING                NET
Year 1:   ▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓ ฿195M   ▓▓▓▓ ฿3M                +฿192M
Year 2:   (no new income)          ▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓
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# Year 1 Funding Request Breakdown

| Phase | When | Prisons | Ask | Cost | Deliverables |
|---|---|---|---|---|---|
| **Phase 1: Pilot** | Month 1-3 | 1 | ฿40M | ฿0.8M | 1 prison live |
| **Phase 2: Expand** | Month 4-6 | +1 | ฿40M | ฿0.6M | 2 prisons live |
| **Phase 3: Scale** | Month 7-9 | +2 | ฿60M | ฿0.7M | 4 prisons live |
| **Phase 4: Complete** | Month 10-12 | +2 | ฿55M | ฿0.52M | 6 prisons live |
| **Year 1 Total** | | **6** | **฿195M** | **฿2.62M** | |

**Year 1 Efficiency:** ฿2.62M spent / ฿195M received = **98.7% preserved**

# Year 1: Phased Deployment

## Phase 1: Pilot (Month 1-3)

**Goal:** Deploy to 1 prison, validate technology, train staff

**Ask:** ฿40M | **Cost:** ฿800,000

| Item | Cost |
|---|---|
| 11 PCs (1 prison) | ฿110,000 |
| Cloud (3 months) | ฿450,000 |
| Deploy + Train | ฿150,000 |
| Contractor | ฿90,000 |

**Deliverables:** 1 prison live, 11 PCs, video calls working, AI moderation, staff trained

**Success:** 100+ calls, 99%+ uptime

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## Phase 2: Expand (Month 4-6)

**Goal:** Deploy to 1 more prison, load test, security audit

**Ask:** ฿40M | **Cost:** ฿600,000

| Item | Cost |
|---|---|
| 11 PCs (1 prison) | ฿110,000 |
| Cloud (3 months) | ฿340,000 |
| Load Testing | ฿50,000 |
| Security Audit | ฿100,000 |

**Deliverables:** 2 prisons live (cumulative), 22 PCs, load tested at 200+ concurrent, security audit passed

**Success:** 200+ concurrent calls, 99.9% uptime, 0 critical bugs

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## Phase 3: Scale (Month 7-9)

**Goal:** Deploy to 2 more prisons, optimize performance

**Ask:** ฿60M | **Cost:** ฿700,000

| Item | Cost |
|---|---|
| 22 PCs (2 prisons) | ฿220,000 |
| Cloud (3 months) | ฿400,000 |
| Deploy + Train | ฿80,000 |

**Deliverables:** 4 prisons live (cumulative), 44 PCs, optimized system

**Success:** 300+ weekly calls, 99.5%+ uptime

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## Phase 4: Complete (Month 10-12)

**Goal:** Deploy to remaining 2 prisons, full production

**Ask:** ฿55M | **Cost:** ฿520,000

| Item | Cost |
|---|---|
| 18 PCs (2 prisons) | ฿180,000 |
| Cloud (3 months) | ฿250,000 |
| GPU Inference | ฿90,000 |

**Deliverables:** 6 prisons live (all), 62 PCs, 500+ calls/week, 24/7 monitoring

**Success:** 99.5%+ uptime, 80%+ staff satisfaction

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## Year 1 Summary

| Phase | Prisons | PCs | Ask | Cost |
|---|---|---|---|---|
| Phase 1 | 1 | 11 | ฿40M | ฿0.8M |
| Phase 2 | +1 (2 total) | +11 (22 total) | ฿40M | ฿0.6M |
| Phase 3 | +2 (4 total) | +22 (44 total) | ฿60M | ฿0.7M |
| Phase 4 | +2 (6 total) | +18 (62 total) | ฿55M | ฿0.52M |
| **Total** | **6** | **62** | **฿195M** | **฿2.62M** |

**Deployment Progress:**

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Phase 1: [▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓] 1/6 prisons (17%)
Phase 2: [▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓] 2/6 prisons (33%)
Phase 3: [▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓] 4/6 prisons (67%)
Phase 4: [▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓] 6/6 prisons (100%)
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## Year 1 Cost-Saving Strategies

| Strategy | Details | Pros | Cons (Compliance Impact) |
|---|---|---|---|
| **Low-Spec Cloud** | Use minimal cloud resources vs over-provisioned | Lower cloud costs | ⚠️ **User experience:** May affect uptime, slower loading times, lower satisfaction |
| **AI Sampling Mode** | 1 frame/30sec vs 30fps real-time | 98% GPU cost savings | ❌ **PDPA violation:** Insufficient monitoring may miss data breaches |
| **Cloud Infrastructure** | Rent VMs vs buy servers | No capex, flexible scaling | ❌ **PDPA violation:** Data may leave Thailand without proper safeguards |
| **Cloud GPU** | Cloud GPU vs owned hardware | No GPU purchase needed | ❌ **Cybersecurity Act violation:** Third-party processing without Thai compliance |
| **Minimal Team** | Contractors vs full-time employees | No benefits, flexible | ⚠️ **PDPA risk:** No dedicated DPO (Data Protection Officer) |
| **No Office** | Remote work only | Save ฿1.2M+ setup cost | No direct compliance impact |
| **Deferred ISO27001** | Start certification Year 2 | Save ฿3M+ Year 1 | ❌ **ISO27001 not certified:** Cannot prove security standards to government |
| **Deferred PDPA** | Full compliance Year 2 | Focus on deployment | ❌ **PDPA violation:** No consent management, no data inventory, no DPO |
| **Deferred Cybersecurity Act** | Start Year 2 | Save compliance costs | ❌ **Cybersecurity Act violation:** No incident |

| Strategy | Details | Pros | Cons (Compliance Impact) |
|---|---|---|---|
| | | | response plan, no security measures documented |
| **PC Bulk Purchase** | Buy 62 PCs at once, negotiate discount | Lower unit cost | No direct compliance impact |
| **Domestic Cloud** | Use Thai cloud provider | Free domestic bandwidth | ⚠️ **PDPA risk:** GPU inference still requires international data transfer |

## Compliance Risk Summary

| Law | Year 1 Status | Risk Level | When Fixed |
|---|---|---|---|
| PDPA | ❌ Non-compliant | High | Year 2 |
| ISO27001 | ❌ Not certified | Medium | Year 3 |
| Cybersecurity Act | ❌ Non-compliant | High | Year 2 |
| ETDA | ❌ Non-compliant | Medium | Year 2 |

## Cost Impact Summary

| Strategy | Year 1 Savings | Compliance Risk |
|---|---|---|
| Low-Spec Cloud | Lower cost | Low (user experience) |
| AI Sampling | ฿1.5M+ | Medium |
| Cloud vs Owned | ฿30M+ | High |
| Contractors vs Full-time | ฿5M+ | Medium |
| No Office | ฿1.2M+ | None |
| Deferred Compliance | ฿5M+ | High |
| **Total Year 1 Savings** | **฿40M+** | |

## What We're NOT Cutting

| Area | Why Not Cut |
|---|---|
| Hardware Quality | PCs must be reliable for prison environment |
| Security Basics | Basic security practices still required |
| Staff Training | Officers must be properly trained |
| System Monitoring | 24/7 monitoring essential for operations |
| Call Recording | Required for compliance and review |

# Years 2-5: Thai Sovereignty Investment

## Year 2: Build Infrastructure

**Goal:** Buy all servers, achieve PDPA compliance, hire Thai team

**Cost:** ฿43M

| Category | Amount |
|---|---|
| Infrastructure (22 servers, 2 GPU stacks, 100TB storage, network) | ฿32.9M |
| Compliance (PDPA + Cybersecurity + ETDA + ISO27001 start + audit) | ฿6.5M |
| Team (5 engineers) + Office | ฿6M |

**Key Deliverables:** Own all hardware, data in Thailand only, PDPA compliant

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## Years 3-5: Operations & Scaling

**Goal:** ISO27001 certification, scale team, stable operations

| Year | Cost | Team | Focus |
|---|---|---|---|
| Year 3 | ฿16.1M | 10 engineers | ISO27001 certification |
| Year 4 | ฿16.1M | 13 engineers | Stable operations |
| Year 5 | ฿16.1M | 13 engineers | Feature development |

**Total Years 2-5:** ฿91.3M

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## Infrastructure Summary (Year 2)

We own everything after Year 1. No more cloud rentals.

| Type | Count | Purpose | Cost |
|---|---|---|---|
| Video Servers | 4 | LiveKit routing (400+ concurrent) | ฿3.2M |
| App Servers | 4 | Agent workers, API | ฿2M |
| Database Servers | 4 | PostgreSQL + Redis | ฿1.2M |
| Support Servers | 6 | Load balancers, monitoring, backup | ฿1M |
| GPU Stacks | 2 | AI moderation (8 GPUs total) | ฿20M |
| Storage | 2 | 100TB NVMe | ฿3M |
| Network + DC | - | Switches, racks, power | ฿2.5M |
| **Total** | **22** | | **฿32.9M** |

## Team Growth

| Year | Engineers | Cost | Type |
|---|---|---|---|
| Year 1 | 3-4 | ฿0.5M | Founders + contractors |
| Year 2 | 5 | ฿4.8M | Start full-time hires |
| Year 3 | 10 | ฿9.6M | Scale team |
| Year 4-5 | 13 | ฿12.5M/yr | Full team |

**5-Year Team Cost:** ฿36M (38% of budget)

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## 5-Year Financial Summary

| Year | Ask | Spend | Cumulative | Remaining |
|---|---|---|---|---|
| Year 1 | ฿195M | ฿3M | ฿3M | ฿192M |
| Year 2 | - | ฿43M | ฿46M | ฿149M |
| Year 3 | - | ฿16.1M | ฿62.1M | ฿132.9M |
| Year 4 | - | ฿16.1M | ฿78.2M | ฿116.8M |
| Year 5 | - | ฿16.1M | ฿94.3M | ฿100.7M |
| **Total** | **฿195M** | **฿94.3M** | | **฿100.7M** |

**Margin:** ฿100.7M (52%)

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## Spending by Category

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Infrastructure:  35% (฿32.9M)
Team:            38% (฿36M)
Compliance:      13% (฿12.5M)
Operations:       5% (฿4.5M)
Cloud Y1:         2% (฿1.7M)
Hardware:         1% (฿0.6M)
Contingency:      6% (฿6.1M)
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## Cloud vs Owned Infrastructure

| Factor | Year 1 (Cloud) | Year 2+ (Owned) |
|---|---|---|
| **PDPA Compliance** | ❌ Data may leave Thailand | ✅ Data stays in Thailand |
| **Cost (5 years)** | ฿25M+ (renting) | ฿32.9M (owned once) |
| **Latency** | ~50ms | ~5ms |
| **Government Trust** | ❌ Foreign cloud | ✅ Thai-owned |
| **AI Moderation** | ⚠️ Sampling (1 frame/30sec) | ✅ Full real-time (30fps) |
| **Control** | ❌ Vendor dependent | ✅ Full control |

## Risk Mitigation

| Risk | How We Handle It |
|---|---|
| Year 1 delivery fails | Phase gates - funding released only on success |
| GPU prices spike | Buy in Year 2, have ฿101M buffer |
| Compliance delayed | Start Year 2, hire consultants |
| Can't hire engineers | Competitive salaries, remote options |
| AI misses violations (Year 1) | Manual review + call recording |
| Year 1 compliance violation | Document risk, fix immediately in Year 2 |

## Key Assumptions

- Operating hours: 9am-4pm (6 hours/day)
- Call frequency: 2 meetings/month/prisoner, 20 min each
- Peak concurrent calls: ~100
- Thai senior engineer: ฿80k-120k/month
- Bangkok office: ฿100k/month
- Colocation: ฿50k/rack/month

## Summary

| Question | Answer |
|---|---|
| How much do you need? | ฿195M in Year 1 (one-time) |
| How much will Year 1 cost? | <฿3M (ultra lean) |
| What if it fails? | Phase gates - funding released only on success |
| How is Year 1 so cheap? | Cloud + contractors + deferred compliance |
| What about Years 2-5? | Own 22 servers + 2 GPU stacks, Thai team, full compliance |
| What's the total project cost? | ~฿94M over 5 years |
| What's the margin? | ฿101M (52%) |
| Is it compliant? | Year 1: No (calculated risk). Year 2+: Yes |
| Is data in Thailand? | Year 1: Partially. Year 2+: Yes (owned infrastructure) |
| What's the compliance risk? | High in Year 1, resolved by Year 2 with dedicated budget |